

GARLIC - Genomic Autozygosity Regions
Likelihood-based Inference and Classification v1.1.6
User Manual

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September 20, 2026

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1 Introduction

Extended tracts of homozygosity in individual genomes manifest as a result of haplotypes inherited identical by descent (IBD) from both biological parents. These runs of homozygosity (ROH) are important genomic features, with their length distributions and genomic locations informative about population history and useful for the mapping of recessive loci contributing to Mendelian and complex diseases. However, precisely because the length distribution of ROH is affected by population history, a simple one-size-fits-all genotype-counting approach to inferring ROH in multiple populations is ill-advised. Here, we present a model-based method and accompanying software package for inferring ROH in microarray-derived SNP genotype that incorporates population-specific parameters and genotyping error rates as described in Pemberton *et al.* (2012). Version 1.1.0 includes several updates, including a method for reweighting LOD scores by gaps between SNPs, the option to report ROH lengths in cM, and the ability to provide per-genotype error rates.

If you use GARLIC please cite the following articles,
ZA Szpiech, A Blant, TJ Pemberton. (2017) GARLIC: Genomic Autozygosity Regions Likelihood-based Inference and Classification. Bioinformatics doi: 10.1093/bioinformatics/btx102.

TJ Pemberton, et al. (2012) Genomic patterns of homozygosity in worldwide human populations. American Journal of Human Genetics, 91, 275 – 292.

2 Obtaining GARLIC

NOTE: In order to successfully run, windows users must have `ann.figtree.version.dll` and `figtree.dll` (provided) in the same folder as `garlic.exe`.

GARLIC pre-built binaries and source code are available at <https://github.com/szpiech/garlic>. To compile from source, change directories to the `src/` directory and type `make`; the Makefile detects the platform from `uname`, so no editing is needed. GARLIC's only external dependency is the zlib library (<http://www.zlib.net/>), provided by the system on Linux and macOS, where the Makefile simply links `-lz`. Threading uses `std::thread` from the C++11 standard library rather than calling pthreads directly; libstdc++ implements it over pthreads on Linux, so `-pthread` is still passed there, but no pthreads library is named or vendored on any platform. GNU GSL and FIGTree were required by earlier versions and are no longer used: FIGTree was replaced by a direct Gaussian sum, and GSL by `src/garlic-math.cpp`, which reproduces the routines GARLIC used bit for bit. On Windows, zlib likewise comes from the MinGW toolchain (`mingw-w64-x86_64-zlib` under MSYS2's MINGW64); the 32-bit zlib that earlier versions vendored under `lib/win32` has been removed, since a 64-bit MinGW cannot link it. The windows version of GARLIC is built in an MSYS2/MinGW environment (<https://www.msys2.org/>); since the move to `std::thread` it requires a MinGW GCC configured with the POSIX threading model, which is what provides `std::thread` on that platform.

3 Basic Usage

GARLIC can be run by executing the following command.

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --auto-winsize --out example
```

The example data is derived from human genotype chip data with hg18 coordinates (`--build hg18`), and a priori we do not know how large to make our window so we use the built-in window size selection algo-

rithm (`--auto-winsize`). All output files will be named starting with 'example' (`--out example`). This produces the following files:

```
example.error
example.log
example.60SNPs.kde
example.freq.gz
example.roh.bed
```

If the automatic window size selection algorithm fails, you can output the KDEs of the LOD score distribution for multiple window sizes (without calling ROH) by using the `--winsize-multi` argument, i.e.

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize-multi
30 40 50 60 70 80 90 --out example
```

This will generate KDEs for your inspection. Once you've chosen a window size, you should rerun garlic specifying that size. For example, if you choose a window size of 60 SNPs, then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example
```

If you already know what LOD score cutoff to use (say you are analyzing more individuals from a previously studied population), you can use the `--lod-cutoff` argument, i.e. if your LOD score cutoff is known to be 2.5 then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example --lod-cutoff 2.5
```

If you already know what size thresholds to use for size classification (say you are analyzing more individuals from a previously studied population), you can use the `--size-bounds` argument, i.e. if your size thresholds are known to be 500000 and 1000000 for the boundaries between short/med and med/long, respectively, then you would run

```
garlic --tped example.tped.gz --tfam example.tfam --build hg18 --error 0.001 --winsize 60
--out example --lod-cutoff 2.5 --size-bounds 500000 1000000
```

To run the wLOD, you must provide a map file and give the `--weighted` flag:

```
garlic --tped example.tped.gz --tfam example.tfam --map example.map.gz --weighted --build
hg18 --winsize 60 --out example --error 0.001
```

Other command line arguments are described below.

4 Statistics implemented

We previously advocated the application of a logarithm of the odds (LOD) score measure of autozygosity applied in a sliding-window framework to infer ROH in high-density SNP genotype data (Pemberton *et al.*, 2012). Fundamentally, for SNP k in individual i , this method calculates the log-likelihood ratio

of observed genotype $G_{i,k}$ under the hypotheses of autozygosity and non-autozygosity, incorporating an assumed genotype error rate ϵ and population-specific allele frequencies. The LOD score of window w in individual i is then calculated as the sum of the log-likelihood ratios of the K SNPs in the window:

$$LOD(w, i) = \sum_{k=1}^K \log_{10} \left(\frac{Pr[G_{i,k}|X_k = 1]}{Pr[G_{i,k}|X_k = 0]} \right). \quad (1)$$

Here, $Pr[G_{i,k}|X_k = 1]$ is the probability of observing genotype $G_{i,k}$ under the hypothesis of autozygosity ($X_k = 1$), and $Pr[G_{i,k}|X_k = 0]$ is the probability of observing genotype $G_{i,k}$ under the hypothesis of non-autozygosity ($X_k = 0$). For a biallelic locus with alleles A and B that have population frequencies p_A and p_B and a genotype error rate ϵ , the genotype probabilities under the autozygosity and non-autozygosity hypotheses are given in Table 1.

Calculating $LOD(w, i)$ for all windows in all individuals in a given sample set, examination of the distribution of scores shows clear bimodality, with windows in the left-hand mode supporting the hypothesis of non-autozygosity and those in the right-hand mode supporting the hypothesis of autozygosity. The area under the autozygous mode decreases with increasing window size until it disappears, potentially reflecting the window size beyond which window length is often longer than ROH length leading to the inclusion of non-autozygous regions in the $LOD(w, i)$ calculation that mask the presence of autozygosity. A logical window size to use for ROH detection is therefore the largest window size where the distribution of $LOD(w, i)$ is bimodal, with windows defined as autozygous if their $LOD(w, i)$ is greater than the local minimum between the two modes. Contiguous autozygous windows are subsequently joined to define ROH.

If the weighted flag is used, wLOD scores will be calculated as

$$wLOD(w, i) = \sum_{k=1}^K \log_{10} \left(\frac{Pr[G_{i,k}|X_k = 1]}{Pr[G_{i,k}|X_k = 0]} \right) \times Corr(p_k, [p_1, p_K]) \quad (2)$$

$$\times Pr[norecombination|[g_{k-1}, g_k]] \quad (3)$$

$$\times Pr[nomutation|\mu, [p_{k-1}, p_k]], \quad (4)$$

where

$$Corr(p_k, [p_1, p_K]) = \frac{1}{\sum_1^K LD_{k,i}}, \quad (5)$$

$$Pr[norecombination|[g_{k-1}, g_k]] = e^{-2M(g_k - g_{k-1})}, \quad (6)$$

and

$$Pr[nomutation|\mu, [p_{k-1}, p_k]] = e^{-2M\mu(p_k - p_{k-1})}. \quad (7)$$

$LD_{k,i}$ is computed as r^2 if data are phased and as HR^2 if data are unphased. g_i and p_i are the genetic distance and physical distance at marker i , respectively.

Table 1: Probability model for genotypes under autozygosity and non-autozygosity.

Observed genotype G_k	$Pr[G_{j,k} X_k = 1]$	$Pr[G_{j,k} X_k = 0]$
AA	$(1 - \epsilon)p_A + \epsilon p_A^2$	p_A^2
AB	$2\epsilon p_A p_B$	$2p_A p_B$
BB	$(1 - \epsilon)p_B + \epsilon p_B^2$	p_B^2
Missing	1	1

5 Program Options

Using the command line flag `--help`, will print a help dialog with a summary of each command line option. The full reference is reproduced in Appendix A, which is generated from those same help strings by `make docs` so that it cannot drift from the program.

5.1 Input Files

Genotypes are read from either a VCF (`--vcf`) or a TPED/TFAM pair (`--tped/--tfam`, see <http://pngu.mgh.harvard.edu/~purcell/plink/data.shtml#tr>). The two are mutually exclusive. Either may be read gzipped (<http://www.gzip.org/>) without decompressing first, and a VCF may also be BGZF, as written by `bgzip` or `bcftools`. Consecutive loci are assumed to be in order with respect to their physical position, and all records for a chromosome must be contiguous — what `bcftools sort` produces. If the file contains more than one population, each is analysed separately; see Section 5.3. `--pop` overrides the labels, which for a TPED/TFAM pair are otherwise taken from column 1 of the TFAM.

5.1.1 VCF input

Biallelic SNVs carrying a `GT` field are used. Multiallelic sites, indels, symbolic ALT alleles (`*`, `<NON_REF>`) and records with no `GT` in `FORMAT` are skipped, and the counts are written to the log rather than passed over silently. The `FILTER` column is reported either way and only acted on under `--vcf-pass-only`, since it records the variant caller’s thresholds rather than `garlic`’s.

A VCF states things a TPED cannot, and `garlic` checks them: ploidy is per call, so anything other than a diploid genotype is an error naming the site and the sample; phase is per call, so `--phased` requires `|` everywhere rather than silently mis-reading a `/`; and samples are named in the `#CHROM` line, so `--tfam` is not accepted alongside `--vcf` — it could only disagree. Use `--pop` for population labels; omitting it labels every sample `unknown` and warns, because the allele frequencies really are pooled over all of them. With `--pop`, each population is analysed separately; see Section 5.3.

The counted allele is ALT. For the same genotypes a TPED counts whichever allele it saw first, so the frequency column may come out as $1 - f$ relative to a TPED run. This changes no ROH call, and a `.freq.gz` file remains interchangeable between the two readers: `readFreqData` compares the `ALLELE` column against the reader’s own coding and flips the frequency when they differ.

Genetic map positions are not part of a VCF, so `--weighted` and `--cm` still require `--map`.

Per-genotype genotype qualities are read from the `FORMAT` column with `--gl-type`. This is the only correct way to use a VCF’s `PL` or `GL`: a VCF normalises both so the called genotype’s value is exactly 0, so that single value is 0 for every call, confident or not, and the information is in how much worse the alternatives are. `garlic` therefore reads the whole array and computes

$$P(g) \propto 10^{-\text{PL}_g/10}, \quad \text{error} = 1 - \frac{P(\text{called})}{\sum_g P(g)},$$

with `GL` entering the same expression as `PL = -10 GL`. This reproduces the `GQ` that encodes the same confidence: a `PL` of (0, 30, 60) gives 0.001, exactly `GQ` 30. If the requested field is absent the run stops and the message names whichever of `GQ`, `PL` and `GL` the record does carry.

The companion script `vcf2tped.pl` is deprecated. It writes a TPED plus a `tgls` file, and a `tgls` file holds one value per genotype, which cannot represent a normalised `PL` or `GL`; `garlic` now rejects such a file rather than computing with it.

5.1.2 File formats

tped/tfam (required):

<http://zzz.bwh.harvard.edu/plink/data.shtml#tr>

map (required when `-weighted` or `-cm` are set):

<chr> <snpid> <genetic pos> <physical pos>
<http://zzz.bwh.harvard.edu/plink/data.shtml#map>
(negative positions will cause errors)

tgls (optional; legacy — see the note on `vcf2tped.pl` above):

<chr> <snpid> <unused> <physical pos> <GL ind1> <GL ind2> ... <GL indN>

One value per genotype. Under `--gl-type PL` or `GL` these are read as the probability that the genotype is *correct*, which is *not* what a VCF's PL and GL fields hold; a file of VCF-derived values is rejected with an error. `--gl-type GQ` is the one form whose VCF values can be used here directly.

pop (optional; with `--vcf`, the only way to set population labels):

<sample id> <population> [sex]

Matched by sample ID, never by position. Note the column order is the opposite of a TFAM's first two columns; a TFAM-ordered file is diagnosed as such. The optional third column uses PLINK sex coding and only makes the sex-chromosome warning specific. `freq` (optional, output during computation, can be generated without running whole pipeling with `-freq-only`):

CHR SNP POS ALLELE FREQ
<chr> <locus ID> <allele> <freq>

centromere file (required if no `-build` is set):

<chr> <centromere start> <centromere end>

5.1.3 --tped

Use `--tped <string>` to specify a .tped (transposed PLINK; Purcell *et al.* (2007)) file (see <http://zzz.bwh.harvard.edu/plink/data.shtml#tr> for exact specifications) containing genetic variant information.

5.1.4 --tped-missing

Use `--tped-missing <char>` to specify the missing data code in the .tped file. Default is 0.

5.1.5 --tfam

Use `--tfam <string>` to specify a .tfam (transposed PLINK; Purcell *et al.* (2007)) file (see <http://zzz.bwh.harvard.edu/plink/data.shtml#tr> for exact specifications) containing sample information.

5.1.6 --tgls

Use `--tgls <string>` A tgls file containing per-genotype likelihoods. Formatted: `¡chr¡ ¡snpid¡ ¡unused¡ ¡physical pos¡ ¡GL ind1¡ ¡GL ind2¡ ... ¡GL indN¡`. Default: none.

Per-genotype likelihoods should be given as either a phred-scaled (`--gl-type PL`) or a $\log_{10}$ -scaled (`--gl-type GL`) representing the probability that the genotype is correct or a phred-scaled likelihood that the genotype is wrong (`--gl-type GQ`). For example, if $p = 0.999$ is the probability that the genotype is correct, then $PL = -10\log_{10}(p) = 0.00434511774018$, $GL = \log_{10}(p) = -0.000434511774018$, and $GQ = -10\log_{10}(1 - p) = 30$.

5.1.7 --centromere

Use `--centromere <string>` to specify a file containing custom centromere boundaries formatted as `<chr> <start> <end>`

5.1.8 --freq-file

`--freq-file <string>` A file specifying allele frequencies for all variants. File format:

CHR	SNP	POS	ALLELE	FREQ
<chr>	<locus ID>	<pos>	<allele>	<freq>

By default, this is calculated automatically from the provided data.

5.1.9 --map

`--map <string>` Provide a scaffold genetic map, sites that aren't present within this file are interpolated. Sites outside the bounds are filtered. This is required for wLOD calculations and any runs for which you wish to report ROH in units of cM. Formatted: `¡chr¡ ¡snpid¡ ¡genetic pos¡ ¡physical pos¡`. <http://zzz.bwh.harvard.edu/plink/data.shtml#map> (negative positions will cause errors)

5.2 Output Files

`garlic` produces several files as output. The `.log` file will record the runtime parameters and `.error` will record any errors that occur; `.error` is created only if something is actually written to it. A `.params.json` file records the effective value of every flag together with the values resolved during the run (selected window size, LOD cutoff, size class boundaries and random seed), and can be replayed with `--load-params`.

A gzipped `.freq` file will be output giving the allele frequencies for each locus in the dataset. Formatted as

CHR	SNP	POS	ALLELE	FREQ
<chr>	<locus ID>	<pos>	<allele>	<freq>

With more than one population the file carries one frequency column per population, headed by its label:

CHR	SNP	POS	ALLELE	POPA	POPB
<chr>	<locus ID>	<pos>	<allele>	<freq>	<freq>

This is the format `--freq-file` reads, so a run's own frequency file can be given back to `garlic` to reproduce that run. A file with a single frequency column is applied to every population, whatever the

column is called, so a frequency file from an earlier version keeps working. `--freq-only` writes the same file without performing the analysis.

With `--froh`, a `.froh.tsv` table gives each individual's autozygous total and the fraction of the analysed genome it covers, one row per individual per size class plus a row for all classes together. The denominator is stated in the header, as are the size-class boundaries actually used, so the fractions can be recomputed or compared across runs.

Kernel density estimation of the LOD score distribution is output to `.<winsize>SNPs.kde` formatted as

```
<LOD score> <density>
```

ROH calls are output to `.roh.bed`, which are formatted as UCSC BED files. There is one track per individual (all individuals in the same file) and the data are formatted as

```
<chr> <ROH start> <ROH end> <size class> <ROH length> <placeholder> <placeholder> <placeholder> <RGB track color>
```

Following the BED specification, `<ROH start>` is 0-based and `<ROH end>` is exclusive, so `<ROH end>` minus `<ROH start>` equals `<ROH length>`. Note that under `--cm` the boundaries remain physical positions while `<ROH length>` is reported in cM, so that identity does not hold in that mode. Versions before this one wrote `<ROH start>` as a 1-based position, making every tract appear one base short in a genome browser; coordinates from those versions are shifted by one relative to these.

When requested, raw LOD scores are also output. This file contains one row per individual (corresponding to the order in the `.tfam` file), and LOD scores for each window are given on the columns.

```
<LOD score window 1> <LOD score window 2> ... <LOD score window N>
```

5.2.1 --out

Use `--out <string>` to provide a base name for an output file. This will be used in place of `<outfile>` above. Default value is `outfile`.

5.3 Multiple populations

The LOD score is parameterised by population allele frequency, so pooling distinct populations inflates heterozygosity relative to any one of them and biases the ratio for all of them. When the input contains more than one population label, `garlic` therefore analyses each population separately and in turn. Each receives

- allele frequencies computed from its own individuals;
- its own site filtering, so a site that is monomorphic in one population but polymorphic in another is dropped where it is uninformative;
- its own automatically selected window size, LOD score cutoff and size-class boundaries;
- its own random number stream, derived from `--seed`, so its result does not depend on how many populations preceded it in the file.

A value given explicitly on the command line is not re-derived, so `--winsize`, `--overlap-frac`, `--lod-cutoff` and `--size-bounds` apply to every population.

The consequence worth stating plainly is that analysing several populations together gives each of them exactly what analysing it alone would give. The populations do not interact; running them in one command is a convenience, not a joint model.

Populations are processed one at a time rather than together, so peak memory is bounded by the largest population rather than by their sum.

5.3.1 Where the labels come from

For a TPED/TFAM pair the label is column 1 of the TFAM. **PLINK** calls that column the family ID, and in many files it is one family — sometimes one sample — per value. Such a file will produce one analysis per family, which is almost certainly not what was intended. **garlic** warns when the smallest population has fewer than five individuals, and again when there are more than two populations per individual on average, but it cannot tell the two situations apart with certainty. Use `--pop` to supply real population labels, or `--pool-populations` to ignore them.

A VCF carries no population labels, so `--pop` is the only way to set them; without it every sample is labelled **unknown** and the analysis is pooled.

5.3.2 Output files

With one population the output file names are unchanged. With more than one, the population label is inserted before the extension: `<out>.<POP>.roh.bed`, `<out>.<POP>.froh.tsv`, `<out>.<POP>.params.json` and `<out>.<POP>.<winsize>SNPs.kde`. The `.log`, `.error` and `.freq.gz` files are written once for the whole run.

5.3.3 `--pool-populations`

This flag makes **garlic** ignore the labels and analyse every individual together, which is what it did before per-population analysis existed. Allele frequencies, the LOD score cutoff, the window size and the size classes are then computed over every sample at once, and the unlabelled output names are restored. Use it when the labels are not populations, or to reproduce a run from an earlier version. The `.froh.tsv` header records that pooling happened, and so does `.params.json`.

5.3.4 When a population cannot be analysed

A small population can fail size-class fitting or cutoff selection where a pooled fit would have succeeded. **garlic** reports which population failed and stops, rather than writing a partial set of results that looks complete. Supplying `--size-bounds` or `--lod-cutoff` avoids the automatic step that failed.

5.4 Sex chromosomes

A hemizygous genotype is written as a homozygous call, so a sex chromosome analysed as an autosome is called as one run spanning its whole length in every heterogametic individual, and any F_{ROH} computed from that is inflated. **garlic** therefore refuses to analyse a chromosome that looks sex-linked until it is told what to do with it.

5.4.1 Declaring the system

`--sex-system` takes `xy`, `zw` or `none`. It supplies the one thing no amount of genotype data can: which sex *code* is heterogametic. PLINK codes 1 male and 2 female whatever the species' sex determination system is, so in a ZW species the *homogametic* sex is the one coded 1. Leaving the flag unset is not the same as `none`: unset means nothing has been said and the run stops, while `none` asserts that every chromosome is diploid in every individual.

The conventional names follow from the system — X and Y under `xy`, Z and W under `zw`, plus M or MT for the mitochondrion. A chromosome named anything else is declared explicitly with `--sex-chr` (diploid in the homogametic sex), `--sex-chr-degenerate` (carried only by the heterogametic sex) or `--haploid-chr`. A chromosome's role does not depend on the system: X and Z are both the shared chromosome, Y and W both the degenerate one. Only the shared chromosome is analysed; the others cannot carry a run of homozygosity in anybody and are dropped with a log line.

The numbers 23, 24, 25 and 26 are X, Y, the pseudoautosomal region and the mitochondrion under PLINK's coding of *human* data, and ordinary autosomes in any species with that many chromosomes. `garlic` resolves them only when `--build` names a human assembly, which is already the user asserting human coordinates; without one it refuses rather than picking a reading.

5.4.2 What the analysis does

Allele frequencies on the shared sex chromosome are computed over *everyone*, with the heterogametic sex contributing one allele and the homogametic sex two. Runs of homozygosity are then called only in the individuals who are diploid there. The LOD score cutoff, the window size and the size-class boundaries are estimated on the autosomes and applied to the sex chromosome; `garlic` reports what the sex chromosome's own windows would have given, so the assumption is visible. This follows Cotter *et al.* (2024) and, for the cutoff, the reason they give: one chromosome's windows from part of a cohort usually cannot locate a second minimum in the density.

That is the default rather than the only option. A cohort where the reported diagnostic shows the autosomal cutoff is badly placed on the sex chromosome can be given another with `--sexchr-lod-cutoff`, and a different window size with `--sexchr-winsize`; both apply to the shared sex chromosome alone and leave every autosomal call exactly where it was. They are values you supply rather than a switch asking `garlic` to re-estimate, because the reason the autosomal cutoff is the default is that the estimate on that chromosome usually fails — the diagnostic is where the value comes from. A LOD score cutoff belongs to the window size it was estimated at, so `--sexchr-winsize` requires `--sexchr-lod-cutoff`: the autosomal cutoff no longer describes windows of another size. Both are recorded in `<out>.params.json` as `sexchr_winsize` and `sexchr_lod_cutoff`, which hold the values that were applied whether or not the flags were given, so the record says what happened without a reader having to reconstruct it from the command line.

A half call carries no genotype, which is what the rest of the pipeline needs to know about it. The LOD score takes its default branch for one, so `--tgls` and a VCF's GQ, PL or GL need no special handling — the per-genotype error rate is never read there. That last point is load-bearing for a hemizygous VCF: a haploid PL array holds one value per allele, two at a biallelic site rather than the three a diploid site has, and it is never parsed. Both LD statistics behind `--weighted` skip an uncalled genotype too, so the LD matrix on the shared sex chromosome is estimated among the individuals who are diploid there; under `--ld-subsample` the subsample for that chromosome is drawn from those individuals, so the number requested is the number that informs it.

Who is diploid there is established by a check that always runs. Each individual's heterozygosity on that chromosome is compared against `--het-rate-bounds` (0.02 and 0.10 by default): below the first it reads as hemizygous, above the second as diploid, and between them as ambiguous, in which case the individual

is neither called there nor counted towards the allele frequency. A recorded sex is checked against this and an unrecorded one is inferred from it. The per-individual result is written to `<out>.sexcheck.tsv`. Heterozygous calls in an individual with one copy of the chromosome are impossible by construction, so they are counted and discarded; if more than half the individuals with a recorded sex contradict it, **garlic** stops, because that means either `--sex-system` is inverted or the chromosome was called as diploid — and the alternative is discarding every one of those calls silently.

F_{ROH} is reported separately for the autosomes and for the sex chromosome, each with its own denominator, and is **NA** rather than zero for an individual that cannot be autozygous there. The two are not summed: the X coalesces faster than the autosomes and carries more homozygosity at the same level of consanguinity, so they are quantities to compare rather than to pool.

5.4.3 Pseudoautosomal regions

A pseudoautosomal region is diploid in both sexes. `--par` takes one or more `<chr>:<start>-<end>` intervals and `--par-file` reads them from a file; both accept any number of regions per chromosome — humans have two — and the two flags accumulate. Unlike `--centromere`, repeated rows for one chromosome add a region rather than replacing one. The loci inside are dropped, each region’s count is reported separately, and a region that matches no locus warns by name.

They are identified by coordinates on a declared sex chromosome and never by a chromosome code, because PLINK’s 25 is an ordinary autosome in most species.

For the human X, `--build` supplies them: the coordinates are the Genome Reference Consortium’s own definition for that assembly, recorded with their source in `centromeres/par_regions.txt`, and they are filed under whatever the data calls that chromosome — `chrX` in one file and 23 in the next. Naming any region with `--par` replaces them, and `--par none` keeps them all. No table is supplied for a shared sex chromosome that is not the human X, since a species’ own coordinates are not the assembly’s.

Running without them — on a non-human assembly, or under `--par none` — has a bounded cost, which is worth knowing rather than discovering. The heterogametic sex’s genotypes in the pseudoautosomal region are recoded as hemizygous along with the rest of the chromosome, so their real heterozygous calls there are counted as impossible and discarded — inflating `n_het_recoded` by roughly that region’s share of the markers, about 1–2% of the human X — and their contribution to the allele frequency in it is biased, because dropping heterozygotes while keeping homozygotes is not a random thinning. Both are bounded by the total pseudoautosomal share of the chromosome, which is the quantity that grows in a species that has several.

5.4.4 Chromosome names

Names are matched case-insensitively and ignoring a leading `chr`, so `chrX`, `chrX`, `CHRX`, `X` and `x` are one chromosome, and so are 1, 01 and `chr1`. Output keeps whatever the input said. Two chromosomes in one file whose names differ only that way are an error, and so is the same name appearing twice, which means one chromosome whose rows are not contiguous — **garlic** starts a new chromosome whenever the chromosome column changes and would otherwise analyse it as two.

5.5 Controlling GARLIC

The remaining flags control window construction, LOD score calculation, cutoff selection, size classification, reproducibility and output. They are listed with their types and defaults in Appendix A, which **make docs** generates from **garlic**’s own help strings. This section previously restated them by hand, which is how it came to document `--auto-winsize-size` — a flag the program has never accepted — and to miss every flag added since.

6 Change Log

A sex chromosome is no longer analysed as an autosome; a run containing one stops until
--sex-system says which sex code is heterogametic. See "Sex chromosomes" under Program Options.
Frequencies there come from everyone with the heterogametic sex contributing one allele, calling
is restricted to the individuals who are diploid, the cutoff and size classes are estimated on
the autosomes, and FROH is reported for the two regions separately with NA where an individual
cannot be autozygous. The FROH table is now one row per individual. --autosomes-only drops every
non-autosomal role rather than four names, and 23-26 are resolved only under a human --build.
--par excludes pseudoautosomal regions by coordinate, any number of them.
A hemizygous VCF is
readable on a declared sex chromosome. Chromosome names match case-insensitively and ignoring a
chr prefix.

UNRELEASED - Input with more than one population label is now analysed one population at a
time; see "Multiple populations" under Program Options. Each population gets allele frequencies from its own individuals, its
own site filtering, its own automatically selected window size, LOD cutoff and size-class
boundaries, and its own random stream. A value given on the command line is not re-derived
and applies to every population. Analysing several populations together gives each of them
what analysing it alone gives.

With ONE population nothing changes. With more than one, the label is inserted before the
extension: <out>.<POP>.roh.bed and so on. --pool-populations restores the previous behaviour
exactly, and its output is byte-identical to the previous version's.

Column 1 of a TFAM is garlic's population label but PLINK's FAMILY ID, so a file that uses it
per family will now produce one analysis per family. garlic warns, but use --pop or
--pool-populations.

--freq-file accepts one frequency column per population, matched by header; a single-column

file still applies to everything. The .freq.gz garlic writes, from the analysis or from --freq-only, can be given straight back with --freq-file.

24AUG2017 - v1.1.6 Modified behavior of --cm flag based on user feedback. When this flag is set ROH lengths and length classifications will be reported in cM but ROH boundaries will be reported in bps.

18JUL2017 - v1.1.5 Introduced --ld-subsample argument. If set > 0, will use only a subset of individuals for computing LD statistics when calculating wLOD scores.

18MAY2017 - v1.1.4 If a freq file is provided, GARLIC now ensures that internal coding of alleles is consistent with the allele specified in the file. Improved handling of loci with 100% missing data. Improved reading of gzipped freq files. Improved lod score cutoff discovery. Removed default setting for --winsize, which must now be specified. Now allows --auto-winsize to be used with --winsize-multi, will choose best from the list.

12MAY2017 - v1.1.3 adds a new option for --gl-type called GQ, which is a phred-scaled probability that the genotype is wrong. If $p = 0.999$ is the probability that the genotype is correct, then $GQ = -10 \cdot \log_{10}(1-p) = 30$. The default for --gl-type is now set to none, so if you provide a TGLS file you will be required to choose between GQ, PL, and GL.

08MAY2017 - v1.1.2b fixes a bug that causes a crash when --overlap-frac is set to 0. Setting to zero now chooses the smallest sensible overlap fraction (1/winsize). If you wish to let garlic automatically guess overlap fraction, use the new command line option: --auto-overlap-frac. Also includes an iteration counter to monitor progress of GMM portion of the code.

07MAY2017 - v1.1.2 fixes a bug that causes a crash when a tgls file indicates a genotype has 0 likelihood. Add progress bars for LD calculations and LOD score calculations. Improve efficiency of tgls file loading.

06MAY2017 - v1.1.1a fixes a bug introduced with 1.1.1 that caused crashed when --resample is used.

05MAY2017 - Update to version 1.1.1. For some large datasets (e.g. WGS), KDE can take extremely long to complete, even when only a subset of individuals are considered. Here we introduce some changes to speed this calculation up. Instead of passing all LOD score windows to the KDE function, we now pass only non-overlapping windows. This reduces the number of points sent

to the function by a factor of WINSIZE. If you wish to retain all LOD scores, you may set the `--no-kde-thinning` flag. We also change the default number of random individuals for KDE (`--kde-subsample`) to 20, which still may be set to 0 to use all individuals.

We also introduce a lower bound on the size of ROH reported. Now, an ROH will not be called unless it contains at least `WINSIZE*OVERLAP_FRAC` number of SNPs. For example, for a WINSIZE (`--winsize`) of 100 and an OVERLAP_FRAC (`--overlap-frac`) of 0.25 the smallest possible ROH that will be reported will be 25 SNPs long.

Finally, if `--overlap-frac` is set to 0, GARLIC will attempt to guess a good choice.

02MAY2017 - Update to version 1.1.0a. Bug fixed that caused crashes when using TGLS file.

28APR2017 - Update to version 1.1.0. Includes several updates, including a method for reweighting LOD scores by gaps between SNPs, the option to report ROH lengths in cM, and the ability to provide per-genotype error rates.

25OCT2016 - Update to version 1.0.1. This update fixes a bug where ROH that extended upto the end of the chromosome failed to get assembled and reported. This update also introduces a new command-line flag `--overlap-frac` which is designed to reduce false positive calls by requiring a SNP be covered by at least OVERLAP_FRAC (default 0.25) proportion of high scoring windows to be included in an ROH call. This helps to reduce false positive calls near the boundaries of true ROH.

05MAY2016 - Initial release of GARLIC v1.0.0

A Command line reference

Generated from `garlic`'s own help strings by `make docs` in `src/`; do not edit `flags.tex` by hand.

`--M <int>` The expected number of meioses since a recent common ancestor for `--weighted` calculation.
Default: 7

`--auto-overlap-coef <double1> ... <doubleN> <slope> <intercept>`: coefficients of $\text{overlap\%} = \text{slope} * \log(\text{density}) + \text{intercept}$, used by `--auto-overlap-frac`. The defaults (6.375 63.888) are an empirical fit to human SNP-array data.
Default: 6.375 63.888

`--auto-overlap-frac <bool>` If set, GARLIC attempts to guess based on marker density.
Default: false

`--auto-winsize <bool>` If `--weighted` is set, guesses the best window size based on SNP density, otherwise initiates an ad hoc method for automatically selecting the # of SNPs in which to calculate LOD scores. Starts at the value specified by `--winsize` and increases by `<step size>` SNPs until

finished.

Default: false

--auto-winsize-coef <double1> ... <doubleN> <slope> <intercept>: coefficients of `winsize = slope*log(density) + intercept`, used by **--auto-winsize** with **--weighted**. The defaults (8.3235 138.0521) are an empirical fit to human SNP-array data; override them for other ascertainment schemes or species.

Default: 8.3235 138.0521

--auto-winsize-step <int> Step size for automatic window selection algorithm.

Default: 10

--auto-winsize-threshold <double> Smoothness criterion at which the **--auto-winsize** search stops. Lower values demand a smoother LOD score density and therefore a larger window.

Default: 0.5

--autosomes-only <bool> Drop every chromosome that is not an autosome before calling: X, Y, Z, W and the mitochondrion, recognised by name, plus 23, 24, 25 and 26 when **--build** names a human assembly. A hemizygous genotype is written as a homozygous call and is indistinguishable from true autozygosity, so a hemizygous chromosome is otherwise called as one run spanning its whole length. Does not resolve bare 23-26 without a human **--build**: dropping them is right if they are PLINK's human codes and destroys real autosomes if they are not, so **garlic** refuses instead. See **--sex-system**.

Default: false

--build <string> Choose which genome build to use for centromere locations: hg18, hg19, hg38 or t2t-chm13 (T2T-CHM13v2.0, UCSC hs1).

Default: none

--centromere <string> Provide custom centromere boundaries. Format <chr> <start> <end>.

Default: none

--chr <string1> ... <stringN> Analyse only these chromosomes, e.g. **--chr** chr1 chr2 chrX. Names are matched after normalisation, so '1' and 'chr1' are the same. It is an error to name a chromosome that is not in the data.

Default: all chromosomes in the input

--cm <bool> Measure ROH lengths in genetic distance units. This requires a mapfile.

Default: false

--dump-docs <string> Write the command line reference to stdout in a documentation format and exit: txt for the README block, tex for a LaTeX description list. Used by 'make docs' so the documentation cannot drift from the program.

Default: _none

--error <double> The assumed genotyping error rate.

Default: -1

--force <bool> Overwrite existing output files. Without this, **garlic** refuses to clobber an existing <out>.roh.bed.

Default: false

--freq-file <string> A file specifying allele frequencies for each population for all variants. File format: CHR SNP POS ALLELE FREQ <chr> <locus ID> <allele> <freq> By default, this is calculated automatically from the provided data.
Default: none

--freq-only <bool> If set, calculates a freq file from provided data and then exits. Uses minimal RAM.
Default: false

--froh <bool> Also write <out>.froh.tsv: per-individual autozygous total and fraction, broken down by ROH size class. The denominator is stated in the file header.
Default: false

--gl-type <string> Which genotype-quality field to use. With **--vcf** it is read from the FORMAT column; with **--tgls** it names the form of the single value per genotype in the .tgls file. The two sources are NOT equivalent. With **--vcf** (recommended): GQ error = $10^{-(GQ/10)}$, the VCF spec's definition directly. PL The whole PL array is read and converted to a posterior: $P(g)$ proportional to $10^{-(PL_g/10)}$, error = $1 - P(\text{called})/\text{sum}$. This is the only correct use of a VCF's PL. A VCF normalises PL so the CALLED genotype is exactly 0, so that one value carries no information – it is 0 for every call, confident or not. The information is in how much worse the alternatives are. GL The same, via $PL = -10 * GL$. A missing field is an error naming whichever of GQ/PL/GL the file does carry, and a called genotype with '.' for the field is an error rather than a silent fall back to **--error**. With **--tgls** (legacy; the format holds ONE value per genotype): GQ As above. The only form a VCF's values can be dropped into. PL error = $1 - 10^{-(PL/10)}$, i.e. $P(\text{genotype CORRECT})$. *** NOT the VCF PL field. *** Under this definition PL=0 means an error of 0, and the integers map to 0, 0.21, 0.37, 0.50, ... – none a realistic error rate (0.001 would be PL=0.00435). Feeding VCF PLs here is rejected. Use **--vcf --gl-type PL** instead. GL error = $1 - 10^{GL}$, with the same caveat.
Default: none

--gmm-max-iter <int> Maximum EM iterations when fitting ROH size classes.
Default: 1000

--gmm-tol <double> Convergence tolerance for the ROH size-class EM fit.
Default: 1e-05

--haploid-chr <string1> ... <stringN> Chromosomes haploid in every individual: mitochondrion, chloroplast. Dropped before calling, since a run of homozygosity on them is not a meaningful quantity.
Default: names matching M or MT, plus 26 under a human --build

--help <bool> Prints this help dialog.
Default: false

--het-rate-bounds <double1> ... <doubleN> <lo> <hi>: heterozygosity rates on the shared sex chromosome below which an individual reads as hemizygous and above which it reads as diploid. Between them it is ambiguous, and garlci neither calls ROH there for that individual nor counts its alleles. Used to check every recorded sex and to infer the ones that were not recorded. Fixed bounds rather than a two-component fit on purpose: a cohort that is entirely one sex has no bimodality to find, and a clustering rule would invent a split there. These fail the same way for every dataset.
Default: 0.020000

--kde-cut <double> Extend the density grid this many bandwidths beyond the range of the data.
Default: 3

--kde-points <int> Number of grid points at which the LOD score density is evaluated. Also sets the resolution at which the between-mode minimum (the LOD cutoff) is located.
Default: 512

--kde-subsample <int> The number of individuals to randomly sample for LOD score KDE. If there are fewer individuals in the population all are used. The default, 0, uses every individual, which makes the selected LOD cutoff a deterministic function of the data alone. A positive value subsamples, which saves memory but makes the cutoff depend on **--seed**.
Default: 0

--kde-thin-step <int> Take every Nth LOD score window when building the KDE. 0 (the default) uses the window size, which keeps the sampled windows non-overlapping; 1 uses every window. Replaces **--no-kde-thinning**, which is equivalent to 1 and is kept as a deprecated alias.
Default: 0

--ld-subsample <int> The number of individuals to randomly sample for LD calculation during wLOD. If there are fewer individuals in the population all are used. Set ≤ 0 to use all individuals (will increase runtime).
Default: 0

--load-params <string> Read flag values from a <out>.params.json written by a previous run. Flags given on the command line take precedence over the file, so a run can be repeated with one parameter changed.
*Default: **--none***

--lod-cutoff <double> For LOD based ROH calling, specify a single LOD score cutoff above which ROH are called in all populations. By default, this is chosen automatically with KDE.
Default: -999999

--map <string> Provide a scaffold genetic map, sites that aren't present within this file are interpolated. Sites outside the bounds are filtered. This is required for wLOD calculations and any runs for which you wish to report ROH in units of cM.
Default: none

--max-gap <int> A LOD score window is not calculated if the gap (in bps) between two loci is greater than this value.
Default: 200000

--max-winsize <int> Upper bound on the window size that **--auto-winsize** will try. The search previously had no bound and would grow past the number of loci if the smoothness criterion was never met.
Default: 1000

--mode-smooth-span <int> Number of adjacent grid points used to smooth the density when locating its modes. Must be smaller than **--kde-points**.
Default: 20

--mu <double> Mutation rate per bp per generation for **--weighted** calculation.
Default: 1e-09

- nclust** <int> Set number of clusters for GMM classification of ROH lengths.
Default: 3
- no-centromere** <bool> Treat every chromosome as having no assembled gap, instead of requiring **--build** or **--centromere**. For organisms with no centromere gap in the assembly, and for references garlic has no built-in table for (only hg18, hg19 and hg38 are built in; T2T-CHM13 is not). Mutually exclusive with **--build** and **--centromere**.
Default: false
- no-kde-thinning** <bool> Set this flag to send all LOD score data to KDE function. This may dramatically increase runtime.
Default: false
- out** <string> The base name for all output files.
Default: outfile
- outdir** <string> Write all output files into this directory, which is created if it does not exist. Equivalent to prefixing **--out** with the path.
- overlap-frac** <double> The minimum fraction of overlapping windows above the LOD cutoff required to begin constructing a run. ROH will have a lower bound size threshold of $WIN_SIZE \times OVERLAP_FRAC$. If set to 0, GARLIC sets the value to the lowest sensible value: $1/winsize$.
Default: 0.25
- par** <string1> ... <stringN> Pseudoautosomal regions on the shared sex chromosome, as <chr>:<start>-<end>, comma or space separated. They are diploid in both sexes and too short to call at a useful window size, so the loci inside them are dropped. ANY NUMBER of regions per chromosome: humans have two, other species have more, and a system with several shared sex chromosomes can have them on each. Coordinates are read in the assembly the input uses; there is deliberately no way to name a pseudoautosomal region by chromosome code, because PLINK's 25 is an ordinary autosome in most species. **--build** supplies the human X regions on its own, from the Genome Reference Consortium's definition for that assembly; see centromeres/par_regions.txt. Naming any region here replaces them, and **--par none** keeps them all.
Default: the regions for --build, or none without one
- par-file** <string> The same regions from a file, one per line, <chr> <start> <end>. '#' comments and blank lines are ignored and .gz is accepted. Unlike **--centromere**, repeated rows for a chromosome ADD a region rather than replacing one. May be given together with **--par**; the regions accumulate.
Default: none
- phased** <bool> Set if data are phased and you want to calculate r^2 instead of h^2 while **--weighted** is set. Uses extra RAM. Has no effect on computations without **--weighted**.
Default: false
- pool-populations** <bool> Analyse every individual together as one population, ignoring the population labels. This is what garlic did before per-population analysis, and it restores the old output names as well: <out>.roh.bed rather than <out>.<POP>.roh.bed. Allele frequencies are then computed by pooling everyone, which inflates heterozygosity relative to any one population and biases the LOD scores for all of them. Use it when the labels are not populations – column 1 of a TFAM is the PLINK family ID, which is often one family or even one sample per value – or to reproduce an

earlier run.

Default: false

--pop <string> A file mapping sample ID to population, replacing the population labels that would otherwise come from the TFAM. Whitespace separated, '#' comments and blank lines ignored, .gz accepted: <sample_id> <population> [sex] Two required columns in THAT order. Note this is the opposite order to a TFAM, whose first column is the population. The optional third column is sex in PLINK coding (1 male, 2 female, 0 or -9 unknown); unlike a TFAM, anything else is an error rather than being read as unknown. Rows are matched by sample ID, never by file order. Every sample in the data must have a row; extra rows are ignored and counted, so a cohort-wide file can be used against a subset.

Default: none

--quiet <bool> Suppress the progress bar and informational messages. Errors and warnings still go to stderr and to <out>.error.

Default: false

--raw-lod <bool> If set, LOD scores will be output to gzip compressed files.

Default: false

--resample <int> Number of resamples for estimating allele frequencies. When set to 0 (default), garlic will use allele frequencies as calculated from the data.

Default: 0

--seed <int> Random seed, for reproducible runs. Affects **--kde-subsample**, **--ld-subsample** and **--resample**. If 0 (default), a nondeterministic seed is drawn and written to the log file so the run can be repeated exactly.

Default: 0

--sex-chr <string1> ... <stringN> The chromosome diploid in the homogametic sex and hemizygous in the heterogametic one: X under xy, Z under zw. Only needed when the name is not the conventional one, e.g. **--sex-system zw --sex-chr LG12**. Requires **--sex-system**.

Default: the conventional name for --sex-system

--sex-chr-degenerate <string1> ... <stringN> The chromosome carried only by the heterogametic sex: Y under xy, W under zw. No individual can carry a run of homozygosity on it – hemizygous in one sex and absent in the other – so it is dropped before calling.

Default: the conventional name for --sex-system

--sex-system <string> Which sex is heterogametic, and therefore which individuals are hemizygous on the shared sex chromosome: xy individuals coded 1 (male) in the TFAM or **--pop** are heterogametic zw individuals coded 2 (female) are heterogametic none assert that every chromosome is diploid in every individual, and stop reporting sex chromosomes entirely PLINK sex coding is 1 male and 2 female whatever the species' sex determination system is, so in a ZW species the HOMOGAMETIC sex is coded 1. This is the one thing no amount of genotype data can supply, which is why it is a flag. It is not an analysis path: it fills in a per-chromosome, per-individual expected ploidy that one code path then reads. Leaving it unset is NOT the same as 'none'. If a sex chromosome is present and this is unset, garlic stops rather than analysing it as an autosome.

Default: unset

- sexchr-lod-cutoff** <double> Call the shared sex chromosome at this LOD score cutoff instead of the one estimated on the autosomes. Every run logs what the sex chromosome's own windows would have given, which is where the value for this comes from. The autosomal cutoff is the default because one chromosome's windows from part of a cohort often cannot locate a second minimum in the density at all; that is also why this is a value you supply rather than a switch that asks garlic to re-estimate.
Default: the autosomal cutoff
- sexchr-winsize** <int> Build windows of this many SNPs on the shared sex chromosome instead of the size used on the autosomes. The overlap FRACTION is unchanged, so the number of windows a SNP must fall in scales with the window. A LOD score cutoff belongs to the window size it was estimated at, so with this the autosomal cutoff no longer applies to the sex chromosome; give **--sexchr-lod-cutoff** as well.
Default: the autosomal window size
- size-bounds** <double1> ... <doubleN> Specify the size class boundaries ROH boundaries. By default, this is chosen automatically with a 3-component GMM. Must provide numbers in increasing order.
Default: -1.000000
- tfam** <string> A tfam formatted file containing population and individual IDs.
Default: none
- tgls** <string> A tgls file containing one per-genotype likelihood value per individual, in the same row order as the tped. See **--gl-type** for what the values must mean: for PL and GL this is NOT the VCF convention.
Default: none
- threads** <int> The number of threads to use. Applies to LOD score calculation, the LD calculations under **--weighted**, the LOD score KDE, and ROH assembly.
Default: 1
- tped** <string> A tped formatted file containing map and genotype information.
Default: none
- tped-missing** <char> Single character missing data code for TPED files.
Default: 0
- vcf** <string> A VCF file (plain or gzipped) to read genotypes from, instead of **--tped**. Biallelic SNVs with a GT field are used; multiallelic sites, indels and symbolic ALTs are skipped and the counts reported. Sample names come from the #CHROM line, so **--tfam** is neither needed nor accepted; use **--pop** to supply population labels. Sites must be grouped by chromosome. Ploidy must be 2 at every call, and with **--phased** every call must use '—'.
Default: none
- vcf-pass-only** <bool> Skip sites whose FILTER column is neither PASS nor '.'. Without this, such sites are KEPT and their number is reported, because a FILTER column reflects the caller's thresholds rather than garlic's.
Default: false
- verbose** <bool> Show the progress bar even when stderr is not a terminal. By default it is drawn only on a terminal, because it works by emitting backspaces.
Default: false

`--version` <bool> Print the version and exit.

Default: false

`--weighted` <bool> Compute LOD scores weighted by LD and probability of mutation.

Default: false

`--winsize` <int> The window size in # of SNPs in which to calculate LOD scores.

Default: 0

`--winsize-multi` <int1> ... <intN> Provide several window sizes (in # of SNPs) to calculate LOD scores. LOD score KDEs for each window size will be output for inspection.

Default: -1

`-h` <bool> Prints this help dialog.

Default: false

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